OCTANE™

Power Desktop Workstation



Breakthrough System Architecture for High-Performance Visual Computing

Pure Performance

OCTANE from Silicon Graphics is the high-performance workstation designed to power the next generation of visual computing solutions. OCTANE lets users take control of large, complex, and continually growing data sets that would normally stall conventional computer systems. Whether in manufacturing, entertainment, visual simulation, defense imaging, or the sciences, OCTANE brings interaction to complex computing tasks with its revolutionary new architecture. This interaction is the key to innovation and insight.

Switching away from Tradition

OCTANE employs a radical new architecture that shatters the bottlenecks associated with conventional systems. OCTANE replaces the traditional "one-at-a-time" shared bus with a "one-to-one" crossbar switch. This switch allows different computer subsystems to communicate directly, without interfering or competing with other system activity.

Graphics Leadership

OCTANE offers the most powerful desktop graphics available, providing unmatched realism and polygon performance. The system is highly modular and scalable, allowing users to choose from three graphics configurations to suit their current and future needs.

For applications that require additional space or two separate screens, OCTANE supports a dual-head option. OCTANE/SI and OCTANE/SSI are ideal for superior solid modeling, and OCTANE/MXI adds to the outstanding solid model performance of OCTANE/SSI graphics by providing unrivaled texture for 3D graphics.

The Power of Symmetric Multiprocessing

With OCTANE, Silicon Graphics now delivers symmetric multiprocessing to the desktop. OCTANE provides the option for one or two MIPS® R10000™ processors that operate simultaneously or independently on separate tasks. The threaded 64-bit operating system of OCTANE, combined with threaded applications, further boost the system's computational capabilities so that concurrent engineering design and analysis applications will benefit.

Robust Digital Media

The OCTANE Personal Video option allows users to create and manipulate content for a variety of uses ranging from collaboration to videoconferencing to multimedia Web sites. The OCTANE Compression option provides state-of-the-art compression capabilities that can be used for anything from Web-based movie-making to high-end broadcast graphics. To satisfy the demanding requirements of video professionals, OCTANE Digital Video™ delivers two streams of uncompressed real-time video and video texturing for unique special effects.

Beyond PCI

The OCTANE system has a modular, expandable design that supports a wide range of peripherals and I/O devices. The base system includes built-in 10Mb/100Mb Ethernet™, plus two UltraSCSI buses. OCTANE also supports three optional PCI-64 slots, offering connections to a wide range of industry-standard peripherals. To expand beyond the capabilities of PCI-based devices, OCTANE provides four high-speed XIO™ slots, which have data rates of 1.6 gigabytes per second (peak), 1.2 gigabytes per second (sustained).

OCTANE

Technical Specifications

PROCESSOR DATA

Microprocessor	MIPS RISC R10000 64-bit CPU
Processors	1 to 2
Primary cache	32KB two-way set-associative on-chip instruction cache, 32KB two-way set-associative on-chip data cache

SYSTEM CONFIGURATION

CPU capacity	1 to 2 R10000 @ 175MHz 1 to 2 R10000 @ 195MHz
Memory capacity	64MB to 2GB ECC protection
I/O bandwidth	1.6GB/sec/port (6 ports)
XIO capacity	4 board slots
PCI capacity	1 half-height, 2 full-height board slots (with PCI card cage option)
Storage controllers	2 Ultra SCSI, 1 internal SE, 1 external SE (40MB/sec)
Internal storage capacity	3 3.5-inch drive bays (27GB)
Communication	10Base-T/100Base-TX Ethernet, 2 serial RS422/RS423, (DB-9) 1 bidirectional parallel, 6 audio I/O ports
Texture cache	4MB standard on OCTANE/MXI, optional on OCTANE/SI, OCTANE/SSI

GRAPHICS

Advanced features	Alpha blending, accumulation buffer, anti-aliased RGB lines and points, texture mapping, fog, lighting features (spot lighting, eight light sources, two-sided lighting, ambient, diffused, and specular), arbitrary clipping planes, depth cueing, soft shadow and depth of field, sub-pixel positioning, stenciling, stereo graphics, pan and zoom, X11 pixel operations
Color maps	1 (4,096 colors each)
IRIS GL™/OpenGL	RGBA quad buffer (stereo)
Display modes	RGBA double buffer, RGBA single buffer, color index double buffer, color index single buffer, color index overlay

AUDIO/VIDEO

Analog input/output	Mono-microphone, self-powered stereo desktop loudspeakers with headphone output, stereo analog -10dBV line-level (18-bit A to D and D to A)
Digital input/output	AES3/IEC958 stereo 24-bit digital, optical 24-bit stereo SPDIF, or 24-bit 8-channel ADAT
Jacks and connectors	Microphone: 3.5mm, speaker headphones: 3.5mm, line in/out: RCA, AES3/IEC958: AES-3id/RCA, optical: 12.8Mb/sec EIJA RCZ-6901
Sampling rates	4kHz to 50kHz continuously variable, any analog sample rate locked to any video standard, independent sampling rates for each audio port
Video/compression	OCTANE Digital Video, OCTANE Personal Video, and OCTANE Compression options are available on all OCTANE systems Uncompressed, real-time 8- or 10-bit serial digital I/O, 2 streams of JPEG compression at 2:1, analog video I/O, real-time color space conversion, video texturing, real-time graphics to video out
Genlock	Available on all video options

XIO CARDS SUPPORTED

4-port Ultra SCSI (3 differential, 1 SE or differential),
2-port Fibre Channel*
4-port 100Base-TX and 6 460kbps serial ports
4-port ATM OC3
Serial HIPPI (200MB/sec)

PCI CARDS SUPPORTED

1-port Ultra SCSI
1-port Fibre Channel*
FDDI—single and dual attached and UTP
ATM OC3**
Serial HIPPI**
Token Ring**
ISDN
Second 100Base-TX Ethernet

*available first half 1997
**available through third-party vendors

BUNDLED SOFTWARE

Collaboration	InPerson, WebMagic, OutBox, IRIS InSight, Netscape Navigator, Netscape Mail, Netscape NewsReader, Adobe Acrobat Reader, Cosmo Player, IRIS Showcase
Connectivity	XFS, ISDN/PPP support, TCP/IP, (Novell NetWare™ and Xinet AppleTalk® support optional)
Digital media	Media Player, Media Recorder, Media Convert, Movie Maker, Sound Track, CD/DAT Player, Macromedia Movie Player

PHYSICAL ENVIRONMENT

System	16.25" H x 11.0" W x 13.25" D 14.75" D (depth in localized area of power supply) 16.25" D (depth in localized area of optional PCI module)
20-inch monitor	18.7" H x 18.9" W x 19.9" D; 71.6 lbs.
Voltage and frequency	100-120/200-240 VAC
Heat dissipation	2400BTU/hour
Ambient temperature	+13°C to +35°C operating -10°C to +65°C non-operating
Relative humidity	10% to 80% operating no condensation 10% to 95% non-operating no condensation
Altitude	10,000' operating; 40,000' non-operating
Vibration	0.02", 5-19Hz; 0.35 G, 19-500Hz

REGULATORY AGENCY APPROVALS

Electromagnetic	TBD
Emission	TBD
Product safety	TBD

PROCESSOR BENCHMARKS

	1xR10000 175MHz	1xR10000 195MHz	2xR10000 175MHz	2xR10000 195MHz
AIM	TBD	TBD	700	721
SPEC int95	8.4	9.3	8.5	9.4
SPEC fp95	15.5	17	15.5	17.4
SPECint_base95	7.6	8.5	7.7	8.6
SPECfp_base95	14.2	15.6	14.2	16
SPECint_rate95	N/A	N/A	150	163.5
SPECfp_rate95	N/A	N/A	253	272
LINPACK 1000x1000DP	226	264	226	264
RenderMark	386	432	390	435

GRAPHICS BENCHMARKS

Viewperf Performance Results (frames/sec)

	OCTANE/SI	OCTANE/SI + Texture	OCTANE/SSI	OCTANE/MXI
Advanced visualizer (AWadvs-01)	N/A	10.80	N/A	17.30
CDRS viewset composite (CDRS-03)	26.46	32.29	35.85	48.63
Data explorer viewset composite (DX-03)	7.73	7.73	13.76	13.74
Design review viewset composite (DRV-04)	3.46	4.89	5.43	8.11
Lightscape (Light-01)	1.31	1.31	1.58	1.58
Graphics Primitives				
X lines	3.26M	3.26M	3.26M	3.26M
AA depth cued (fogged) lines	1.28M	1.28M	1.85M	1.85M
Gouraud tri strips*	1.26M	1.26M	2.24M	2.24M
Lit Gouraud Z tri strips*	1.02M	1.02M	1.82M	1.82M
Lit textured Gouraud Z tri strips**	N/A	TBD	N/A	TBD
Lit Gouraud Z independent quads	384K	384K	687K	687K
Gouraud fill (pixels/sec)	120M	120M	240M	240M
Textured Gouraud Z fill (pixels/sec)	N/A	TBD	N/A	TBD
Convolutions 3x3 general (comps/sec)	13.7M	13.7M	24.4M	24.4M
Convolutions 5x5 separable (comps/sec)	11.5M	11.52M	20M	20M
Color lookups (comps/sec)	38M	38M	64M	64M

NOTE: *25 pixel triangles, **50 pixel triangles

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